Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Sunmoon Yu	Email	sunmoon.yu@utoronto.ca
Institute Name	University of Toronto			
Address	184 College Street, Suite 140, Toronto, Ontario, Canada, M5S 3E4			

2. Technology Overview

Title of Proposed Technology	Electrolyte and Interface Engineering to Improve Zinc Metal Battery Calendar Life and Resting Stability			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Mechanical & Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Metal Materials	Energy Materials Technologies
Concept of Proposed Technology	Zinc metal batteries offer a low-cost and safe option for stationary energy storage, but commercialization is limited by self-discharge and capacity loss during resting conditions, which become more severe at elevated temperatures encountered during storage, transportation, and field operation.[1–3] This project aims to improve the calendar life and resting stability of zinc metal batteries by controlling electrolyte solvation and interfacial water structure at the zinc–electrolyte interface. The proposed technology develops practical electrolyte and interface engineering strategies, including optimized electrolyte composition, salt concentration, and targeted additives, to suppress parasitic reactions (e.g., zinc corrosion and hydrogen evolution) under idle conditions. State-of-the-art operando characterization tools will be used to directly probe interfacial reactions and gas evolution under realistic resting and elevated-temperature conditions, enabling quantitative links between solvation structure and self-discharge behavior. By systematically mapping the relationships among solvation structure, temperature (up to 55 °C), and self-discharge rate, the project will deliver electrolyte formulation guidelines that can be implemented without major changes to cell design or manufacturing. The outcome is zinc battery systems with improved storage stability, higher reliability, and reduced deployment risk for commercial applications.			

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Rationale for Joint R&D Project	Reducing self-discharge and improving calendar life in zinc metal batteries requires direct, operando observation of interfacial reactions under realistic resting conditions and elevated temperatures,[2] capabilities that most SMEs do not have in-house due to equipment cost and specialized expertise. The academic team brings state-of-the-art operando characterization techniques (e.g., operando electrochemical mass spectrometry, surface-enhanced infrared absorption spectroscopy, and atomic force microscopy-infrared spectroscopy) to identify how electrolyte composition, concentration, and additives govern interfacial water structure, parasitic corrosion, and gas evolution during storage. The SME partner contributes application-specific requirements, practical electrolyte formulation constraints (cost, safety, supply chain), and validation in industry-relevant cell formats. This joint R&D structure enables rapid translation of operando mechanistic insights into implementable electrolyte and interface engineering strategies, reducing technical risk and accelerating commercialization of zinc metal batteries with improved resting stability.
Key Words	1. Zinc Metal Batteries 2. Calendar life Improvement 3. Self-Discharge Mechanisms 4. Electrolyte and Interface Engineering 5. Operando Characterization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop electrolyte and interface engineering technologies that significantly reduce self-discharge and improve the calendar life of zinc metal batteries under resting conditions, including elevated temperatures. Specifically, **the project aims to (i) identify** electrolyte compositions, concentrations, and additive strategies that suppress parasitic reactions at the zinc–electrolyte interface, and **(ii) translate** operando interfacial insights into practical electrolyte formulation guidelines validated in industry-relevant zinc battery cells. Through this effort, the project advances zinc metal battery technology from laboratory validation (TRL 4) to prototype-level validation under relevant operating and storage conditions (TRL 6).			
R&D Target Specifications	The project targets the following measurable performance improvements in zinc metal battery cells. • **Self-discharge rate reduction:** ≥50% reduction in capacity loss during resting conditions compared to baseline electrolytes • **Shelf-life improvement:** ≤5% capacity loss after 30 days of storage at room temperature (25±2 °C) • **High-temperature storage stability:** ≤20% capacity loss after 30 days of storage at 55 °C • **Corrosion suppression:** ≥50% reduction in hydrogen evolution rate under resting conditions • **Compatibility:** Electrolyte formulations compatible with existing zinc battery cell designs and manufacturing processes • **Cycling performance retention:** ≥80% capacity retention after 300 cycles at 1 C, with no significant degradation compared to baseline electrolytes, evaluated in accordance with KS C IEC 61960.			
R&D Plan	The proposed project will be executed in three integrated phases (3 years), with milestones and deliverables aligned with TRL progression from 4 to 6: **Phase 1. Baseline Assessment and Operando Diagnostics (Year 1, TRL 4)** Baseline zinc metal battery cells (e.g., Zn/MnO_2/system) will be systematically evaluated under resting (idle) conditions across a range of temperatures to quantify self-discharge behavior, zinc corrosion, and gas evolution. Special emphasis will be placed on elevated temperatures relevant to storage and field deployment. Operando electrochemical mass spectrometry (EC-MS) with fully quantitative, sub-second temporal resolution will be employed to directly measure hydrogen evolution rates as a function of temperature, electrolyte composition, salt concentration, and additive chemistry. These measurements will establish baseline degradation signatures and identify dominant parasitic processes governing capacity loss during storage. **Phase 2. Electrolyte and Interface Engineering (Year 2, TRL 5)** Guided by operando diagnostics, we will develop electrolyte and interface engineering strategies to suppress parasitic reactions during resting conditions. The interfacial solvation environment will be systematically tuned by varying electrolyte composition (e.g., structure-making vs. structure-breaking components), salt concentration, and targeted additives. Interface engineering of the electrode may also be selectively explored, where necessary, to further promote ion transport while forming electronically insulating, protective interphases. Candidate formulations will be screened using accelerated aging protocols to quantify self-discharge, corrosion suppression, and high-temperature stability. Throughout optimization, SME deployment constraints, including cost, safety, chemical compatibility with cell components, and supply-chain viability, will be considered to ensure the solutions are practical and industry-ready. **Phase 3. Prototype Validation and Translation (Year 3, TRL 6)** Optimized electrolyte systems will be validated in industry-relevant zinc battery cell formats (e.g., pouch cells) under extended resting and elevated-temperature storage conditions using realistic state-of-charge and temperature profiles. Key metrics, including capacity retention, self-discharge rate, and gas evolution, will be benchmarked against baseline electrolytes and verified across multiple			

	prototype builds for reproducibility. Practical implementation factors (e.g., wetting/impregnation and component compatibility) will also be assessed. Based on these results, electrolyte formulation windows and interface engineering design rules will be finalized to support scale-up, manufacturability, and commercialization by the SME partner.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (University of Toronto) will lead the scientific and technical foundation of the project, focusing on operando diagnostics and electrolyte/interface design. **Key responsibilities include:** • Designing and executing operando characterization experiments (e.g., operando EC-MS and interface-sensitive spectroscopy) to identify self-discharge, corrosion, and gas evolution mechanisms under resting and elevated-temperature conditions. • Developing electrolyte and interface engineering strategies, including systematic variation of electrolyte composition, salt concentration, and additives to control interfacial water structure. • Establishing quantitative structure–performance relationships linking solvation environment to self-discharge behavior. • Providing electrolyte formulation guidelines and design rules based on operando insights to enable translation to industry-relevant systems. • Training graduate researchers and managing data analysis, reporting, and technical documentation to support technology transfer.
Expected R&R of Korean SME Partner	The Korean SME partner will lead application-driven development and validation to ensure commercial relevance. **Key responsibilities include:** • Defining performance targets and validation criteria in alignment with intended commercial use cases. • Evaluating candidate electrolyte formulations using in-house pouch cell assembly lines and cyclers to ensure technical feasibility under realistic resting and elevated-temperature conditions. • Establishing a feedback loop by applying the institute's proposed technologies to the field and providing specific feedback to ensure practical R&D aligned with industrial needs. • Assessing manufacturability, safety, cost, and supply-chain compatibility of proposed electrolyte and additive systems. • Validating performance improvements through prototype testing and reproducibility studies. • Conducting reproducible prototype testing and providing structured feedback to refine electrolyte and interface designs. • Supporting scale-up assessment and ensuring alignment with regulatory and market requirements. • Building a robust IP portfolio by deriving practical and process patents based on the research institute's foundational research to strengthen technology barriers.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project will deliver practical electrolyte and interface engineering solutions that significantly reduce self-discharge and improve the calendar life and resting stability of zinc metal batteries, including under elevated-temperature storage conditions. **Key technical outcomes include** validated electrolyte formulation guidelines and additive strategies that suppress parasitic reactions such as zinc corrosion and hydrogen evolution, enabled by state-of-the-art operando characterization under relevant operating environments. **From a commercialization perspective,** the project will advance zinc metal battery technology to TRL 6 through prototype-level validation in industry-relevant cell formats. The resulting improvements in storage stability, reliability, and thermal robustness will reduce technical risk, improve product consistency, and lower warranty and deployment costs for SME partners. Importantly, the proposed solutions are designed to be directly integrated into existing SME manufacturing infrastructures, enabling rapid adoption without major capital investment. **The broader impact of this project includes** strengthening the competitiveness of Korean SMEs in the secondary battery market, particularly for stationary and backup energy storage applications where safety, cost, and long calendar life are critical. By enabling commercially viable zinc metal batteries, the project supports diversification beyond lithium-ion technologies, enhances supply-chain resilience through the use of abundant materials, and contributes to sustainable energy storage solutions with positive environmental and economic benefits. In addition, the project establishes **a strategic Canada–Korea research and innovation partnership** that leverages complementary strengths, namely, advanced operando electrochemical characterization and interface science from Canada (University of Toronto), and application-driven development and commercialization capabilities from Korean SMEs, creating a pathway for continued joint innovation and follow-on technology translation.

6. References

(1) *Energy Environ. Sci.*, 2019, 12, 3288–3304

(2) *Adv. Energy Mater.* 2026, 16, e03630

(3) *Joule* 2025, 9, 101784

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization